

8(a)	<table><tr><td></td><td>letter</td></tr><tr><td>high concentration of protons</td><td>M ;</td></tr><tr><td>location of photosynthetic pigments</td><td>L or N ;</td></tr><tr><td>site of light-independent stage</td><td>K ;</td></tr><tr><td>site of light-dependent stage</td><td>L or N ;</td></tr></table> <i>ignore M in the last row</i>		letter	high concentration of protons	M ;	location of photosynthetic pigments	L or N ;	site of light-independent stage	K ;	site of light-dependent stage	L or N ;	4
	letter											
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location of photosynthetic pigments	L or N ;											
site of light-independent stage	K ;											
site of light-dependent stage	L or N ;											
8(b)	<i>any four from:</i> 1 any one named ; e.g. chlorophyll b / carotene / xanthophyll / carotenoids ; 2 act as accessory pigments / part of antenna complex / part of light harvesting system ; 3 absorb, light / photons ; 4 pass <u>energy</u> on to, chlorophyll a / primary pigment / reaction centre ; 5 absorb different wavelengths of light / wavelengths not absorbed by chlorophyll a ;	4										

Question	Answer	Marks																																						
8(c)	1 absorption higher for (whole) chloroplasts (throughout) ; 2 comparative data quote or greatest difference at 525 / 530 nm ; <table border="1"> <thead> <tr> <th rowspan="2">wavelength / nm</th><th colspan="2">absorbance / au ±0.05</th></tr> <tr> <th>whole</th><th>pigment</th></tr> </thead> <tbody> <tr><td>500</td><td>9.2</td><td>6.4</td></tr> <tr><td>510</td><td>8.4</td><td>3.8</td></tr> <tr><td>520</td><td>7.4</td><td>2.3</td></tr> <tr><td>525</td><td>7.0</td><td>1.95</td></tr> <tr><td>530</td><td>6.6</td><td>1.85</td></tr> <tr><td>540</td><td>6.2</td><td>2.05</td></tr> <tr><td>550</td><td>5.8</td><td>2.35</td></tr> <tr><td>560</td><td>5.8</td><td>2.7</td></tr> <tr><td>600</td><td>6.65</td><td>4.7</td></tr> <tr><td>650</td><td>8.85</td><td>8.1</td></tr> <tr><td>670</td><td>9.6</td><td>9.2</td></tr> </tbody> </table> 3 (because) pigments arranged for better absorption in chloroplasts / thylakoid membranes are stacked / AW ; 4 (because) chloroplasts contain more pigments ;	wavelength / nm	absorbance / au ±0.05		whole	pigment	500	9.2	6.4	510	8.4	3.8	520	7.4	2.3	525	7.0	1.95	530	6.6	1.85	540	6.2	2.05	550	5.8	2.35	560	5.8	2.7	600	6.65	4.7	650	8.85	8.1	670	9.6	9.2	4
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